

ISE1 Part 2 – Group Activity on Real-World NLP Applications

Overview

Each group of **5 students** will act as a problem-solving team using **Natural Language Processing (NLP)** to address a real-world language problem in one of the following domains:

- Aviation
- Law
- Healthcare
- Education

You will be assigned a real-world scenario (e.g., decoding cockpit conversations, legal contract disputes, analyzing patient feedback, or grading student essays) and are expected to design a complete NLP solution.

Group Task Instructions

1. Understand the Scenario

Identify the language-based challenges from the real-world case assigned to your group.

2. Design an NLP Pipeline

Break the problem into key NLP tasks such as:

- Tokenization
- POS Tagging
- Named Entity Recognition
- Emotion or Sentiment Analysis
- Dialogue Act Classification

Mention the tools/libraries you plan to use (e.g., `spaCy`, `NLTK`, `BERT`, `Whisper`).

3. Draw a Flowchart

Create a visual representation of your NLP pipeline showing how data flows and where major decisions occur.

4. Plan Evaluation

Describe how you will evaluate your system: accuracy, F1-score, manual verification, or comparison with external benchmarks.

5. Present Your Work

Prepare to briefly present your pipeline, analysis, and thought process during in-class group presentations.

Deliverables (Group-wise)

- One-page NLP pipeline diagram on A3 size drawing sheet
- 1–2 page write-up of the problem and NLP approach
- Short group presentation

Submission Deadline

Drawing sheet [**Tuesday, 29th July 2025**]

Presentation [**Wednesday, 30th July 2025**]

Note

This task emphasizes understanding and applying NLP creatively. Focus on problem-solving, collaboration, and communication in your domain—beyond just implementation.

Sample Case – Aviation Domain: Air France Flight 447

Scenario: You are part of a crash investigation team applying NLP to decode communication data from Air France Flight 447. Your job is to design an NLP pipeline to help investigators understand what went wrong by analyzing cockpit recordings.

Objective: To understand how NLP can solve real-world problems by designing a custom pipeline using domain knowledge and creativity.

Solution

Step 1: Understand the Use Case

Brief Summary of the Problem: Investigating pilot communication and cockpit conversations from the black box of Air France Flight 447 to identify warning signs, distress language, or deviations in communication patterns.

Step 2: Decode the Language Challenge

Type of Text Data Involved: Black box CVR transcripts, ATC logs, technical maintenance reports

Key Language Tasks Identified:

- **Word-level:** Identify aviation-specific distress terms ("stall", "pull up", "altitude", "failure"), technical vocabulary, and commands
- **Syntax-level:** Detect incomplete or interrupted sentences, parse subject-verb-object errors under stress, coordination failure markers

Step 3: Design the NLP Pipeline

Main NLP Tasks:

- Tokenization
- POS Tagging
- Named Entity Recognition
- Dependency Parsing
- Emotion Detection
- Dialogue Act Recognition

Tools/Libraries Used:

- Whisper/OpenAI for transcription
- spaCy for linguistic preprocessing
- Transformer models (BERT/Roberta) fine-tuned for emotion analysis

- Dialogue RNN for turn-level analysis

Step 4: Visualize Your Pipeline

Cockpit Audio → Speech-to-Text → Tokenization → POS Tagging → NER → Dependency Parsing → Emotion Detection

Step 5: Evaluation Strategy

How will you evaluate your results?

- Compare NLP findings with official BEA crash investigation
- Measure accuracy of emotional spikes vs timestamps of crisis escalation
- Annotated dataset with critical utterance classification (manual + automated)

Step 6: Final Reflection

What was most challenging?

- Processing low-quality, noisy audio data under extreme stress conditions

What did your team do well?

- Created a multi-layer pipeline combining audio and linguistic context effectively

Any improvements for future design?

- Add multimodal sensor data fusion (altitude, airspeed) with linguistic timeline for full incident reconstruction